KIDSPHONICS / JJCM / 01

# Technical Defense Guide

## Programming language, architecture and system behavior

Para sa JJCM presenters: Taglish explanations ng aktuwal na implementation. Ang guide ay mula sa current source, dependency lockfile, Android configuration at recorded QA/build evidence; hindi ito listahan ng proposed features.

### A 60-second technical introduction

SABIHIN: “Ang pangunahing programming language namin ay Dart at ang UI framework ay Flutter. Provider at ChangeNotifier ang gamit sa shared application state. SharedPreferences ang local storage; naka-JSON ang complex progress snapshots. Local asset files ang images at recorded audio. Sa speech activity, gumagamit kami ng speech_to_text at microphone permission para sa word recognition. Kotlin ang maliit na Android bridge para mabasa ang installed package information.”

### Technical flow at a glance

**Flutter screen/widget receives learner interaction**

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**Screen logic validates the current activity / selected answer**

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**AppProvider updates the appropriate progress model**

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**PreferenceWriteQueue saves the local snapshot**

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**ChangeNotifier updates listening UI / Progress / Parent reports**

| Current scope | Correct explanation |
|---|---|
| Platform | Android APK, portrait-oriented UI; actual device tested by the user. |
| Learning content | 6 lessons, 11 games, 26-letter Quick Check content. |
| Accounts / backend | No custom online login, server, cloud database or teacher portal in the current app. |

Prepared 27 October 2026. App version/build identifiers describe a source baseline, not the number of times the team improved the design.

# 01 Language, framework and packages

| Item | Actual role / version from evidence |
|---|---|
| Dart | Main app language. Recorded build uses Dart 3.11.3; pubspec allows SDK >=3.0.0 <4.0.0. |
| Flutter | UI framework and app build tooling. Recorded corrected build uses Flutter 3.41.5 stable. |
| Kotlin / Android | MainActivity hosts Flutter and exposes installed version/build through MethodChannel. |
| Gradle Kotlin DSL / Android SDK | Android build configuration uses .gradle.kts; Java compatibility and Kotlin JVM target are 17. |
| provider 6.1.5+1 | Dependency access and UI subscriptions to ChangeNotifier state. |
| shared_preferences 2.5.5 | On-device key-value persistence for settings, profile and serialized progress. |
| audioplayers 6.6.0 | Plays bundled recordings, sound effects and background music. |
| speech_to_text 7.4.0 + permission_handler 11.4.0 | Device speech-recognition boundary and microphone-permission handling. |
| crypto 3.0.7 | SHA-256 operations, including the salted local Parent PIN digest. |
| google_fonts 6.3.3 | Typography integration; local font assets are available and runtime fetching is disabled. |
| flutter_test / flutter_lints | Development-time tests and analyzer/lint rules. |
| flutter_launcher_icons 0.14.4 | Development-time generation of Android launcher resources; not a learner feature. |

## Language versus framework

SABIHIN: “Dart ang language na sinusulatan namin. Flutter ang framework na nagbibigay ng widgets, rendering, routing at platform integration. Kotlin ay para sa maliit na native Android integration; hindi Kotlin ang buong app.”

Ang resolved package versions ay mula sa pubspec.lock, hindi lamang sa version ranges sa pubspec.yaml. Ang code editor ay development tool; hindi iyon ang language o runtime.

Code / reference: pubspec.yaml; pubspec.lock; Android build files; recorded evidence/flutter-version.json

# 02 Architecture and responsibility

May hiwalay na screens, reusable widgets, models, data catalogs, provider at services. Ito ay Flutter app na may state/service separation; huwag basta tawaging strict MVC, full Clean Architecture o microservices kung iyon ay hindi ang implemented structure.

| Folder / component | Responsibility |
|---|---|
| lib/screens/ | Page routes and activity-specific interaction: lessons, games, Quick Check, Parents and Settings. |
| lib/widgets/ | Shared presentation, navigation shell, illustrated frames, loading animation and global overlays. |
| lib/models/ | Typed progress/session data and rules: AdventureSession, AdventurePolicy, daily activity and game models. |
| lib/data/ | Explicit lesson/game catalogs, recordings, image mappings and session-order helpers. |
| lib/providers/app_provider.dart | Shared learner state, recommendation, progress aggregation, reward recording and persistence coordination. |
| lib/services/ | Audio, local save queue, screen time, parent authentication, speech and native build information. |
| assets/ + pubspec.yaml | Images, audio and fonts registered for inclusion in the Flutter asset bundle. |

## Example: a completed Quick Check

**Screen collects five first-response results**

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**AdventurePolicy evaluates mastery; AppProvider guards the session ID**

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**Progress / XP / stars / daily activity are updated**

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**JSON snapshot is queued for local saving; subscribed UI refreshes**

Local setState handles short-lived screen interaction. Provider shares state across screens; it does not mean every animation tick must rewrite persistent progress.

Code / reference: lib/main.dart; lib/providers/app_provider.dart; lib/models/letter_adventure.dart; lib/widgets/main_navigation_shell.dart

# 03 Data storage and save flow

## What is the database?

SABIHIN: “SharedPreferences ang storage mechanism namin. Local key-value persistence ito; hindi MySQL, SQLite o Firebase. Ang complex progress structures ay kino-convert sa JSON strings bago i-save.”

| Saved information | Implementation example |
|---|---|
| Authoritative app state | appStateV1: profile, preferences, XP/stars, progress and other app-state values. |
| Letter assessment history | letterAdventuresV1: completed count, best/latest responses, mastery, timestamp and last session identity. |
| Lesson / activity evidence | wordLessonProgressV1, letterProgressV2 and dailyActivityV1. |
| Parent gate | parentPinHashV1 stores the salt and digest together; the authenticated session is not persisted. |
| Screen time | screenTimeStateV1 plus compatible keys for daily usage, limit, extra time and bypass state. |

## Load → validate → update → save

Startup obtains SharedPreferences and decodes the saved state. Typed defaults and validation handle invalid values. A load failure exposes Retry; the app does not intentionally overwrite an unreadable save with an empty learner state.

PreferenceWriteQueue serializes snapshot writes, skips unchanged values, and retries a failed operation up to three times. AppProvider first writes one authoritative appStateV1 JSON value, then mirrors compatibility keys. Unsaved state can expose Retry.

## Important boundaries

• This is not a multi-user transactional server database. The snapshot/save queue reduces inconsistent or out-of-order writes; it is not a guarantee against every hardware/storage failure.

• No custom cloud backup/sync is implemented. Clear Data, uninstall or device loss can remove local progress; OS backup/restore behavior is separate.

Code / reference: AppProvider._loadPrefs / _savePrefs; lib/services/preference_write_queue.dart; lib/models/word_lesson_progress.dart

# 04 Assessment, rewards and recommendation

## Actual mastery rule

PASS = five completed first responses
AND correct count >= 4
AND Letter sounds is correct
AND Sounds in words is correct

| Example | Interpretation |
|---|---|
| 4/5, both sound tasks correct | Passes the implemented mastery policy. |
| 4/5, one sound task wrong | Does not pass; high total alone is insufficient. |
| Incomplete check | Not a completed assessment. Guided retries do not overwrite first-response evidence. |
| Mastered before; lower result later | Latest result can change, while prior mastery and best score remain retained. |

## Duplicate prevention and rewards

AdventureSession has an ID; completeAdventure rejects a repeated completion for the recorded same session ID. It also guards first completion and first mastery. First completed check = 5 XP; first mastery adds 25 XP and 1 star. Repeating mastery does not repeatedly award those milestones.

Games keep their own scoring/reward paths. For example, Flappy completion is not proof of assessed phonics mastery. Replaying a sound or opening Progress does not itself earn mastery.

## Recommendation is rule-based

The current unmastered target remains selected. For another target, prioritize completed-but-unmastered checks, lower latest score and alphabetical order. All mastered → letter review selection. Walang trained recommendation model.

Level = floor(XP / 200) + 1. Streak uses actual dates with scored answers or speaking attempts; speech statistics remain separate from ordinary answer accuracy.

Code / reference: AdventurePolicy.passes; AdventureSession.submit; AppProvider.completeAdventure / recommendedAdventureLetter; docs/PROGRESS_FLOW.md

# 05 Images, audio and animation

## Bundled resources

Images, recordings and fonts are stored under assets/ and declared in pubspec.yaml. Data catalogs map each letter, word or instruction phrase to a local file. The learner does not download the whole lesson library during a normal session.

| Component | What it does |
|---|---|
| PhonicsAudioService | Resolves phrase/letter/word keys to local recordings and coordinates playback. No text-to-speech fallback is claimed. |
| AudioService / background music | Separate sound-effects and music behavior, controlled by saved preferences. |
| Frame animation | Ordered local images are displayed over time. AnimationController/timers and lifecycle checks drive frame changes; this is not generative animation at runtime. |
| Decode / preload controls | precacheImage and ResizeImage reduce repeated loading and request suitable image decode sizes. This does not change the original source image bytes. |
| Lifecycle / accessibility | Covered or backgrounded activities pause relevant motion/playback. Reduced Animations can use a static frame; text scaling has layout support. |

## Why is the APK large?

It includes many local images and recordings, plus Flutter/native libraries for three CPU architectures in the universal package. The recorded corrected APK is 515,866,438 bytes: about 515.9 decimal MB or 492.0 MiB. This is APK file size, not a measurement of runtime RAM or installed app-data usage.

Deleting output/home_classroom removed about 175.6 MB of PC-only exports/previews. Those files were outside the runtime asset bundle, so their deletion did not reduce the APK.

SABIHIN: “Offline availability has a storage trade-off. We can explain the asset footprint honestly; deleting design previews outside the bundle does not optimize the installed app.”

Code / reference: pubspec.yaml; lib/services/phonics_audio_service.dart; lib/widgets/animated_screen_art.dart; lib/widgets/loading_screen_art.dart; APK report

# 06 Speech-recognition technical flow

**Request microphone permission**

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**Initialize device recognizer and choose supported English locale**

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**Listen → receive recognized words and final-result status**

↓

**Normalize words → compare with the expected target**

↓

**Record valid practice result or show retry / support guidance**

## What comparison is actually performed?

matchesSpokenWord lowercases and extracts complete word tokens. Easy/Medium can accept the target as a complete word within a phrase. Hard compares the normalized whole target with the normalized whole recognized result. It does not accept an arbitrary substring inside a different spelling.

Preferred supported English locales are tried in order en_US, en_PH, en_GB, then another supported English locale. The listening request uses a five-second listenFor value and two-second pauseFor value; the device service can still determine actual availability/behavior.

## Limits to explain clearly

• This is word-recognition practice, not phoneme-level accuracy measurement, accent grading, a clinical assessment or a custom-trained AI model.

• Microphone permission and compatible speech-service/language support are required. The app does not force an on-device-only recognition option, so do not guarantee offline speech or that recognition audio never leaves the device.

• No speech / no match / speech timeout can show Retry without counting an empty input as a wrong spoken answer. Permission denial and unavailable service have separate guidance.

Code / reference: lib/services/speech_practice_service.dart; lib/screens/voice_recognition_screen.dart; AndroidManifest.xml

# 07 Parent PIN and daily screen time

| Mechanism | Actual implementation |
|---|---|
| PIN validation | Exactly 4 numeric digits; matching confirmation; rejects 0000, 1111, 1234 and 4321 at setup/change. |
| PIN storage | Random.secure creates a 24-byte salt. The digest is SHA-256 of salt + colon + PIN. Salt and digest are stored together, not the plain PIN. |
| Failed entry limit | Five failed verifications cause a 30-second lockout. The failure/session state is in memory, not a permanent server-side throttle. |
| Authenticated parent session | Not saved as a permanent authenticated login. Leaving the area ends it; lifecycle logic can expire it after extended backgrounding. |
| Daily allowance | A global foreground clock tracks usage; daily limits are 15/30/45/60/90 minutes. Parent-authenticated time is excluded from normal learner counting. |
| Time limit reached | Global TimeLimitOverlay blocks access and pauses covered tickers. A parent can add 10/15/30 minutes or bypass for today. |
| Protected reset | Requires an authenticated parent session and explicit confirmation. It is not part of ordinary startup or normal exit. |

## What security can we claim?

SABIHIN: “May local parental gate kami na may salted PIN hash at temporary lockout. Hindi plaintext ang naka-store na PIN. Pero hindi ito bank-grade authentication o encrypted cloud account system.”

A salted fast SHA-256 digest is not a password-stretching KDF such as Argon2/PBKDF2. SharedPreferences is not encrypted secret storage. A four-digit local gate should be described as parental control within the project scope, not protection against every compromised/rooted-device attack.

ScreenTimeService uses a queued snapshot and periodic checkpoints; date handling is local to the device. Walang remote time-authority service sa current app.

Code / reference: lib/services/parent_auth_service.dart; lib/services/screen_time_service.dart; lib/widgets/time_limit_overlay.dart; lib/screens/parent_screen.dart

# 08 Android package and release build

| Question | Verified technical answer |
|---|---|
| What is delivered? | An Android APK. A universal build includes armeabi-v7a, arm64-v8a and x86_64 native libraries. |
| Minimum Android version? | The recorded corrected candidate declares minSdk 24, meaning Android 7.0 or later. Hardware/service compatibility still needs device testing. |
| Target / compile SDK? | Recorded package metadata says targetSdk 36 and compileSdk 36. The source delegates SDK values to the current Flutter configuration. |
| Package identity? | com.example.kidsphonics. Native MainActivity extends FlutterActivity; the app is portrait-oriented. |
| Native bridge? | MethodChannel kidsphonics/app_info handles getInfo, returning installed package version/build for About information. |
| Release build versus signing? | The build is release mode, but the current Gradle release configuration signs with the existing debug signingConfig. These are different concepts. |
| Public Play Store release? | No publication is established by the recorded local build. A production distribution plan should manage a proper signing key; do not claim this was Play Store-published. |

## Reproducible build outline

Review source and assets
→ run scoped tests / analyzer
→ flutter build apk --release
→ inspect package metadata and signature
→ compare bundled assets with reviewed hashes
→ install and test on the target phone

The recorded corrected build used --no-pub and all three target platforms with existing locked dependencies. For updates, package identity and compatible signing matter; do not promise that every install method preserves data.

Code / reference: android/app/build.gradle.kts; AndroidManifest.xml; MainActivity.kt; lib/services/app_build_info.dart; corrected APK report

# 09 Testing evidence and limitations

| Evidence | How to explain it accurately |
|---|---|
| Recorded full QA | The 26 October report records 1,004 passed / 0 failed in the prior full QA run. This is historical evidence, not a newly repeated full run today. |
| Corrected-build checks | 37 focused regressions passed, plus a separate strict asset check. Flutter analysis reported no issues. |
| Bundled assets | The corrected-build report records 863 declared/runtime assets, including 446 images and 408 audio files, with hash verification. |
| Archive cleanup follow-up | After the user deleted PC-only classroom exports, the obsolete archival test was removed. The remaining seven classroom checks passed. |
| Device testing | The written build report originally marked device testing pending. The user later reported using the APK successfully on their own device. Do not convert this into a claim that all Android models were tested. |

## What the test types mean

• Unit tests: data transformations, assessment policy, matching rules and game-model behavior.

• Widget/regression tests: route behavior, interaction, layout, controls and repeated-action/lifecycle safeguards, often with mocked device services.

• Asset/package checks: file existence, declared assets, SHA-256 equality, APK metadata and signature. These do not prove the educational effectiveness of the content.

## Do not overclaim

Automated success is not a guarantee of zero bugs. Real microphone recognition, audible sound quality, touch responsiveness, low-end memory/performance and accessibility need device observation. Educational improvement requires its own study/user evaluation; the code does not establish research outcomes.

Code / reference: Corrected Universal APK Report; evidence/regressions-summary.json; build/home_classroom_after_output_cleanup.log; user device-test confirmation

# 10 Panel Q&A — implementation

| Question | Short answer to practice |
|---|---|
| 1. Anong programming language? | Dart ang main app language. Flutter ang UI framework; Kotlin ang maliit na native Android bridge. |
| 2. Bakit Flutter? | Our current implementation uses reusable widgets and one Dart UI/state codebase. The tested delivery is Android; cross-platform capability is not proof that we already tested iOS. |
| 3. Ano ang database ninyo? | SharedPreferences local key-value storage, with JSON snapshots for complex progress. Hindi MySQL, SQLite o Firebase ang current implementation. |
| 4. Paano connected ang screens? | Flutter navigation handles routes; AppProvider/ChangeNotifier shares learner state. Screens call app logic/services, and subscribed widgets rebuild. |
| 5. May backend/API server ba? | Walang sariling learner-account backend or cloud database. Native MethodChannel is a Dart-to-Android bridge, not a web API server. |
| 6. Paano hindi nadodoble ang reward? | Assessment completion uses session identity and first-completion/first-mastery conditions. Other games have their own accepted-action and completed-state guards. |
| 7. AI ba ang recommendation? | No trained model: it retains the current target until mastery and selects another using recorded completion, latest score and alphabetical order. |
| 8. Paano ang errors sa saving? | Queued writes and checked return values; retry attempts and an unsaved-state signal. User-facing Retry is available instead of falsely reporting a successful save. |

## A useful answering pattern

Name the technology → explain its role in this app → give a concrete example → state the scope or limitation. Halimbawa: “Provider ang shared state; pag-save ng Quick Check result, nare-refresh ang Progress. Hindi ito cloud synchronization.”

Kapag hindi mo maalala ang exact implementation detail, sabihin kung anong source file ang nagha-handle nito. Huwag manghula ng algorithm, database o external service.

# 11 Panel Q&A — scope and readiness

| Question | Short answer to practice |
|---|---|
| 9. Offline ba lahat? | Local ang content and saved progress. Speech recognition depends on the installed device service, permissions and English support; we do not guarantee offline recognition on every phone. |
| 10. Pronunciation grader ba ito? | No. It compares recognized word tokens with the expected word. It does not measure phoneme quality, accent or clinical speaking ability. |
| 11. Bakit malaki ang APK? | Bundled visuals/recordings and native libraries for three ABIs. APK bytes, installed storage and RAM are different measurements. |
| 12. Paano pinoprotektahan ang Parents? | Four-digit local PIN, random salt plus SHA-256 digest, temporary lockout and nonpersistent authenticated session. It is a local parental gate, not hardened enterprise authentication. |
| 13. Paano nagiging mastered? | Five first-response tasks; at least 4/5 correct plus both sound tasks correct. Game completion and lesson exposure do not independently satisfy this rule. |
| 14. Ano ang mangyayari sa uninstall? | There is no custom cloud recovery. Local data can be lost; OS backup behavior is outside the app-specific storage design. |
| 15. Ready na ba sa defense? | The recorded automated checks passed and the user reported a successful phone test. We still rehearse the exact demo route and keep an APK/video backup. |
| 16. Ano ang puwedeng future work? | Measure educational outcomes, broaden device testing and plan asset-size/distribution improvements. These are future directions, not claims that they are already implemented. |

## Files to remember

Entry/routing: lib/main.dart | Shared state: lib/providers/app_provider.dart | Mastery: lib/models/letter_adventure.dart | Saving: lib/services/preference_write_queue.dart

Speech: lib/services/speech_practice_service.dart | PIN: lib/services/parent_auth_service.dart | Time: lib/services/screen_time_service.dart | Package: android/app/build.gradle.kts + AndroidManifest.xml

Demo companion: KidsPhonics_System_Demo_Flow.docx. Keep answers aligned with the app installed on the presentation phone.